

A Practical Guide to Anomaly detection

TOPIC -- ANOMALY DETECTION

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Explainable anomaly detection Many of the methods discussed above only yield an anomaly score prediction, which often can be explained to users as the point being in a region of low data density (or relatively low density compared to the neighbor's densities). In explainable artificial intelligence, the users demand methods with higher explainability. Some methods allow for more detailed explanations:

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Video surveillance Anomaly detection has become increasingly vital in video surveillance to enhance security and safety. With the advent of deep learning technologies, methods using Convolutional Neural Networks (CNNs) and Simple Recurrent Units (SRUs) have shown significant promise in identifying unusual activities or behaviors in video data. These models can process and analyze extensive video feeds in real-time, recognizing patterns that deviate from the norm, which may indicate potential security threats or safety violations. An important aspect for video surveillance is the development of scalable real-time frameworks. Such pipelines are required for processing multiple video streams with low computational resources.

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Parameter-free Also referred to as frequency-based or counting-based, the simplest non-parametric anomaly detection method is to build a histogram with the training data or a set of known normal instances, and if a test point does not fall in any of the histogram bins mark it as anomalous, or assign an anomaly score to test data based on the height of the bin it falls in. The size of bins are key to the effectiveness of this technique but must be determined by the implementer. A more sophisticated technique uses kernel functions to approximate the distribution of the normal data. Instances in low probability areas of the distribution are then considered anomalies.

-- 04 --

Specific Let T be observations from a univariate Gaussian distribution and O a point from T . Then the z-score for O is greater than a pre-selected threshold if and only if O is an outlier.

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Density Density-based techniques (k-nearest neighbor, local outlier factor, isolation forests, and many more variations of this concept) Subspace-base (SOD), correlation-based (COP) and tensor-based outlier detection for high-dimensional data One-class support vector machines (OCSVM, SVDD)

-- 06 --

Methods Many anomaly detection techniques have been proposed in literature. The performance of methods usually depend on the data sets. For example, some may be suited to detecting local outliers, while others global, and methods have little systematic advantages over another when compared across many data sets. Almost all algorithms also require the setting of non-intuitive parameters critical for performance, and usually unknown before application. Some of the popular techniques are mentioned below and are broken down into categories:

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Petroleum industry Anomaly detection is crucial in the petroleum industry for monitoring critical machinery. Martí et al. used a novel segmentation algorithm to analyze sensor data for real-time anomaly detection. This approach helps promptly identify and address any irregularities in sensor readings, ensuring the reliability and safety of petroleum operations.

-- 08 --

Software ELKI is an open-source Java data mining toolkit that contains several anomaly detection algorithms, as well as index acceleration for them. PyOD is an open-source Python library developed specifically for anomaly detection. scikit-learn is an open-source Python library that contains some algorithms for unsupervised anomaly detection. Wolfram Mathematica provides functionality for unsupervised anomaly detection across multiple data types

-- 09 --

Datasets Anomaly detection benchmark data repository with carefully chosen data sets of the Ludwig-Maximilians-Universität München; Mirror Archived 2022-03-31 at the Wayback Machine at University of São Paulo. ODDS – ODDS: A large collection of publicly available outlier detection datasets with ground truth in different domains. Unsupervised Anomaly Detection Benchmark at Harvard Dataverse: Datasets for Unsupervised Anomaly Detection with ground truth. KMASH Data Repository at Research Data Australia having more than 12,000 anomaly detection datasets with ground truth.

-- 10 --

Ill defined An outlier is an observation which deviates so much from the other observations as to arouse suspicions that it was generated by a different mechanism. Anomalies are instances or collections of data that occur very rarely in the data set and whose features differ significantly from most of the data. An outlier is an observation (or subset of observations) which appears to be inconsistent with the remainder of that set of data. An anomaly is a point or collection of points that is relatively distant from other points in multi-dimensional space of features. Anomalies are patterns in data that do not conform to a well-defined notion of normal behaviour.

-- 11 --

Definition Many attempts have been made in the statistical and computer science communities to define an anomaly. The most prevalent ones include the following, and can be categorised into three groups: those that are ambiguous, those that are specific to a method with pre-defined thresholds usually chosen empirically, and those that are formally defined:

-- 12 --

IT infrastructure In IT infrastructure management, anomaly detection is crucial for ensuring the smooth operation and reliability of services. These are complex systems, composed of many interactive elements and large data quantities, requiring methods to process and reduce this data into a human and machine interpretable format. Techniques like the IT Infrastructure Library (ITIL) and monitoring frameworks are employed to track and manage system performance and user experience. Detected anomalies can help identify and pre-empt potential performance degradations or system failures, thus maintaining productivity and business process effectiveness.

-- 13 --

Quantum Quantum machine learning approaches have been investigated for anomaly detection on near-term quantum hardware. Quantum support vector machines (QSVMs) using quantum kernel methods, such as the ZZFeatureMap, have been applied to classification of normal versus anomalous instances by leveraging high-dimensional quantum feature spaces.

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Intrusion detection Anomaly detection was proposed for intrusion detection systems (IDS) by Dorothy Denning in 1986. Anomaly detection for IDS is normally accomplished with thresholds and statistics, but can also be done with

soft computing, and inductive learning. Types of features proposed by 1999 included profiles of users, workstations, networks, remote hosts, groups of users, and programs based on frequencies, means, variances, covariances, and standard deviations. The counterpart of anomaly detection in intrusion detection is misuse detection.

-- 15 --

Applications Anomaly detection is applicable in a very large number and variety of domains, and is an important subarea of unsupervised machine learning. As such it has applications in cyber-security, intrusion detection, fraud detection, fault detection, system health monitoring, event detection in sensor networks, detecting ecosystem disturbances, defect detection in images using machine vision, medical diagnosis and law enforcement.